Quality Assurance Validating Report

Requirements Engineering Process
Biometric Student Attendance System

Paragon Assurance

19 March 2026

Document ID	Version	Audit Period	Submitted to
PA-QAR-2026-001	1.0	15–19 March 2026	Nevon Solutions (Team P2-1561W)

QA Auditor	Role
Chng Zong Han	QA Auditor

Contents

1 Executive Summary

1.1 Audit Context

Paragon Assurance was engaged by BioSec Educational Institution as an independent external Quality Assurance (QA) vendor to audit the Requirements Engineering (RE) process conducted by Nevon Solutions (Team P2-1561W) for the Biometric Student Attendance System. The audit scope covers three RE methods, Analysing, Specifying, and Validating, in accordance with the QA auditor's mandate. The Eliciting method is excluded, as it was conducted by BioSec prior to vendor engagement and falls outside the QA audit boundary. The audit additionally evaluates the Validating method's QC component by assessing whether the assigned QC team (Veridion Labs) followed a compliant audit process.

The principal process artefacts reviewed are: Analysing Report (NS-AR-2026-001, v1.0), Software Requirements Specification (NS-SRS-2026-001, v0.8), Project Specification (BS-PS-2026-001, v1.0), Requirements repository (GitHub), QC Audit Report (Veridion Labs), and Clarification session records BC-01 (07/02/2026) and TC-02 (09/02/2026).

1.2 Summary of Findings

The audit identified **six (6) process non-compliance instances** across the three RE methods audited. No corrective actions or remediation steps are prescribed; findings are presented to Nevon Solutions for review and decision.

Severity	Count	Error IDs
Critical	1	PA-ERR-002 (SRS Sign-off Absent: Validating Method Incomplete)
High	1	PA-ERR-001 (Open Issues Unresolved)
High-to-Medium	3	PA-ERR-003 (2FA Contradiction), PA-ERR-004 (Enrolment Not Elicited), PA-ERR-005 (Terminology Inconsistency)
Medium	1	PA-ERR-006 (SRS Below Version 1.0)

Table 1: Summary of findings by severity

Key non-compliance areas:

- **The SRS has no customer or vendor signatures anywhere in the document.** All four designated approver fields in SRS §9 are blank. The Validating method requires formal SRS authorisation before external audit release; this step was not completed (PA-ERR-002).
- Three open issues from the Analysing phase remain unresolved in the submitted SRS despite a stated resolution deadline of 20 February 2026 (PA-ERR-001).
- A contradictory 2FA scope specification was not resolved during the Clarify phase (PA-ERR-003).
- The biometric student enrolment process is a prerequisite for all scan-based functions which was not elicited and is absent from the SRS (PA-ERR-004).
- Attendance status terminology is inconsistent across SRS sections and the Data Dictionary (PA-ERR-005).
- The SRS was submitted at Version 0.8, below the Version 1.0 threshold expected for external audit release (PA-ERR-006).

The QC team (Veridion Labs) is assessed as process-compliant: a documented plan with 10 inspection techniques, a self-contained checklist of 21 items, findings covering all three required error types (Authorship, Omission, Contradiction), and a complete sign-off by all four team members. A detailed QA audit of the QC Validating process is presented in Section 6.

2 QA Plan

2.1 Target Process Description

The process under audit is the Requirements Engineering (RE) process as performed by Nevon Solutions for the Biometric Student Attendance System, Customer BioSec Educational Institution, Project Reference P2-1561W. QA audits the *process* that produces the work product, not the work product itself. The error type for all QA findings is **Process Non-Compliance**, an instance where the RE process deviated from defined process requirements through omission, incorrect execution, or unauthorised deviation.

The RE process under audit comprises three sequential methods:

Analysing Method: receiving the BioSec Project Specification, assessing requirements for congruency using the IRC, resolving incongruities through BCIC process (Breakdown, Clarify, Interpret, Categorise), and producing a regulated-revision requirements repository. Phases audited: Breakdown (IRC application, decomposition), Clarify (BC-01 and TC-02 sessions, open issue management), Carrying forward (PS model revision), and Codification.

Specifying Method: interpreting, categorising, and specifying the analysed requirements into the SRS document (NS-SRS-2026-001). Phases audited: Interpret and Categorise, Specify (SRS drafting, template application), iRTM (standalone file, chain-of-custody), and Change mitigation.

Validating Method: applying quality control measures onto the SRS and conducting quality assurance audit. Components: Customer/Vendor validation (SRS sign-off, editorial checks) and QC audit process (plan, checklist, execution, findings preservation).

2.2 Context and Goals

Context: The BioSec project involves developing a cloud-hosted Biometric Student Attendance Management System to replace manual paper-based processes across five campuses in Singapore. The system handles biometric fingerprint data from minors and must comply with Singapore's PDPA. All artefacts were reviewed under a desktop review model.

Goals: (1) Determine whether the Analysing method was applied in full; (2) Determine whether the Specifying method complies with process requirements; (3) Determine whether the Validating method was completed; (4) Identify all instances of process non-compliance with evidence sufficient for review and decision.

2.3 Work Breakdown

The audit was structured in five phases conducted by a single QA auditor (∼8 hours total): Preparation (2h), Evidence Collection (2h), Process Verification (2h), Observation Recording (1h), and Reporting (1h). See Appendix D for the detailed timeline.

3 QA Checklist

The checklist addresses process compliance across four domains using three investigation methods: **M1** (Artefact Inspection), **M2** (Traceability and Cross-Reference), and **M3** (Declaration and Sign-Off Review).
Rating key:
S = Satisfied
PS = Partially Satisfied
NS = Not Satisfied
N/A = Not Applicable.
Full checklist with detailed notes is in Appendix A.

Table 2: QA checklist summary (see Appendix A for full notes)

ID	Checklist Item	Method	Rating
Domain A: Analysing Method			
A.1	PS copy in repository	M1	S
A.2	All PS requirements subjected to IRC analysis	M1	S
A.3	Ambiguous requirements quantified during Clarify	M1, M2	PS
A.4	Clarification sessions with all stakeholder groups	M1	S
A.5	Meeting minutes formally recorded	M1	S
A.6	Compound requirements decomposed	M1, M2	S
A.7	Open issues documented with deadlines	M1	PS
A.8	PS models carried forward	M1, M2	S
A.9	PS models revised during Analysis	M1, M2	S
A.10	IRC exists as standalone document	M1	S
A.11	IRC was actually used	M1	S
A.12	Plan for Analysing phase	M1	S
A.13	PS items codified in repository	M1, M2	S
A.14	Biometric enrolment process elicited	M1, M2	NS
Domain B: Specifying Method			
B.1	SRS references recognised template	M1	S
B.2	Requirements traceable via iRTM	M1, M2	PS
B.3	iRTM exists as standalone file	M1	S
B.4	Terminology consistent across SRS	M2	NS
B.5	Open issues resolved before SRS finalisation	M1, M2	NS
B.6	Contradictions resolved before Specifying	M2	NS
B.7	New model source files present	M1	N/A
B.8	Change Mitigation template exists	M1	S
B.9	SRS at final ($\geq$ v1.0) version	M1	NS
Domain C: Validating Method			
C.1	SRS formally signed off	M3	NS
C.2	Spell check evidence	M3	PS
C.3	Grammar check evidence	M3	PS
C.4	Editorial review evidence	M3	NS
Domain D: Process Artefact Integrity			
D.1	All required artefacts present	M1	S
D.2	Artefacts internally consistent	M2	PS

Table 2: QA checklist summary (see Appendix A for full notes)

ID	Checklist Item	Method	Rating
D.3	AI and Plagiarism declarations present	M1, M3	S
D.4	Repository structured correctly	M1	S
D.5	Change Mitigation Strategies documented	M1	S
D.6	Risk Analysis present	M1	S
D.7	Version control and metadata complete	M1	PS

4 QA Error Detection

All findings are of error type **Process Non-Compliance**.

Error ID	Location	Characteristic
PA-ERR-001	AR §3.5; SRS §2.5, §3.3, §6.5	Three open issues remain unresolved 11+ days past the RPO/RTO deadline of 20/02/2026.
PA-ERR-002	SRS §9; full repository	All four sign-off fields blank. No supplementary authorisation evidence found. Validating method not completed.
PA-ERR-003	SRS §3.1 UR-MOB-06; §5.3 NFR-SEC-03	2FA required for parents/teachers in UR-MOB-06 but restricted to administrators in NFR-SEC-03. Not resolved during Clarify.
PA-ERR-004	SRS §4; AR §3.3–3.4; repository	Biometric enrolment; prerequisite for scan-based operations was not elicited. No requirement exists despite BOR-0-03 referencing it.
PA-ERR-005	SRS §3.1, §4.3.2; Data Dictionary C.4	Three-way terminology inconsistency: “Attended” vs “Excused, Attended...” vs “Present.”
PA-ERR-006	SRS cover; AR cover	SRS at v0.8 (03/03/2026); AR released at v1.0 (08/02/2026). SRS handed over before completion.

Table 3: Error detection register

5 QA Findings

Each finding presents a CRaM analysis of the non-compliance instance. Findings are presented without corrective action; Nevon Solutions is to determine remediation at its discretion. Full evidence tables are in Appendix B.

5.1 PA-FND-001: Open Issues Unresolved at SRS Finalisation

Evidence: See A.1 (Annex)

Error: PA-ERR-001 **Severity:** High **Phase:** Analysing & Specifying

The Analysing Report §3.5 documents three open issues: MCS Integration Security (no deadline), Disaster Recovery RPO/RTO (deadline 20/02/2026), and Biometric Template Storage Format. The SRS dated 03/03/2026 retains all three as unresolved (§2.5, §3.3, §6.5 NFR-DOC-03). The RPO/RTO deadline passed without resolution and no formal deferral was presented. This constitutes a failure of the open issue resolution protocol required before SRS release.

5.2 PA-FND-002: SRS Sign-off Entirely Absent

Evidence: See A.2 (Annex)

Error: PA-ERR-002 **Severity:** Critical **Phase:** Validating

SRS §9 provides sign-off fields for four designated roles; all four rows are entirely blank. The QA auditor conducted a full repository inspection: no signed declaration, no BioSec confirmation document, no interview record, and no alternative authorisation evidence of any kind was found. The Analysing Report sign-off is correctly completed (Loh Wen Xuan, 13/02/2026), confirming the team knows the protocol. The SRS sign-off is the terminal step of the Validating method; its absence means the method was not completed and the SRS was released as an unauthorised draft.

5.3 PA-FND-003: 2FA Scope Contradiction Not Resolved

Evidence: See A.3 (Annex)

Error: PA-ERR-003 **Severity:** High-to-Medium **Phase:** Analysing & Specifying

UR-MOB-06 mandates 2FA for parents and teachers (mobile), while NFR-SEC-03 mandates 2FA exclusively for administrators (web portal). These are directly contradictory. BC-01 and TC-02 session outcomes contain no reference to 2FA scope resolution.

5.4 PA-FND-004: Biometric Student Enrolment Process Not Elicited

Error: PA-ERR-004 **Severity:** High-to-Medium **Phase:** Analysing & Specifying

Biometric enrolment is the process of capturing and registering a student's fingerprint template which is a logical prerequisite for all scan-based operations. No enrolment requirement exists in the SRS (§4), the repository, or the Analysing Report session outcomes. However, BOR-0-03 acceptance criteria reference "biometric enrolment," confirming the team was aware of the concept but did not elicit it.

5.5 PA-FND-005: Attendance Status Terminology Inconsistency

Error: PA-ERR-005 **Severity:** High-to-Medium **Phase:** Specifying

Three-way inconsistency: UR-MOB-01 uses "Attended, Absent, Late, or Excused"; §4.3.2 uses "Excused, Attended, Late, Absent"; Data Dictionary Table C.4 uses "Present." This should have been detected and unified during the Specifying phase.

5.6 PA-FND-006: SRS Submitted Below Final Version Threshold

Evidence: See A.4 (Annex)

Error: PA-ERR-006 **Severity:** Medium **Phase:** Specifying

The SRS bears Version 0.8 (03/03/2026) while the Analysing Report was released at Version 1.0 (08/02/2026). Sub-1.0 version numbers indicate draft status. This is corroborated by incomplete sign-off (PA-FND-002) and unresolved open issues (PA-FND-001).

6 QA Audit of QC Validating Process

This section assesses the QC audit process conducted by Veridion Labs on the SRS. The QA auditor assesses the *process* of the QC audit, not the correctness of individual QC findings. Detailed sub-assessments are in Appendix C.

QC Team: Muhammad Ridwan Putra Jasni (QC Audit Lead), Lin Yuhao, Muhammad Mikhail Bin Mazlan, and Kannan s/o Rajamohan (QC Specialists).

The QC audit is assessed as **process-compliant**:

1. **Plan:** Documented plan with target workproduct, context, goals, 10 inspection techniques (exceed-

ing the minimum of six), role-based work breakdown (4 members), and indicative 4-stage timeline (~6 hours).
2. **Checklist:** Self-contained checklist of 21 items (QC1–QC21) across 7 SRS sections, with CRaM-specific items tailored to the Biometric Student Attendance System.
3. **Error Detection:** 12 error instances across all three required types: Contradiction (3), Omission (5), Authorship (4).
4. **Findings:** 12 findings with Context-Result-Analysis structure and severity classifications (High: 5, Medium: 4, Low: 3).
5. **Reporting:** Retrievable PDF report with structured appendices, cross-referenced error detection, and findings summary.
6. **Sign-off:** All four team members signed.

No QC process non-compliance was identified.

7 Sign-Off

This QA Validating Report (PA-QAR-2026-001, Version 1.0) is formally submitted to Nevon Solutions (Team P2-1561W) by Paragon Assurance. The six (6) process non-compliance findings documented in Sections 4 and 5, together with the QC process audit in Section 6, represent the complete output of the QA audit of Nevon Solutions' Requirements Engineering process for the Biometric Student Attendance System (Project P2-1561W).

The QC audit conducted by Veridion Labs is assessed as process-compliant (Section 6). No QC process non-compliance was identified.

No modifications to any Nevon Solutions process artefact have been made by the QA auditor. No corrective actions or remediation steps are prescribed. This report documents findings only; Nevon Solutions is to determine appropriate action at its discretion.

Role	Name	Signature	Date
QA Auditor	Zong Han	zonghan	19/03/2026

Appendices

A Full QA Checklist

A.1 Domain A: Analysing Method

ID	Checklist Item	Method	Rating	Notes
A.1	Does the vendor have a copy of the Project Specification (PS) in their repository?	M1	S	Project Specification_1561W.pdf present in repository root.
A.2	Were all PS requirements subjected to the Breakdown/IRC analysis (congruency check)?	M1	S	Analysing Report §2.1–2.2 documents full IRC application across all PS requirements. Five incongruity categories assessed.
A.3	Were all ambiguous requirements quantified during the Clarify phase?	M1, M2	PS	Most ambiguous requirements were resolved (BC-01 outcomes). BOR-1-MOB-01 still references “institutional policies” without specific policy documents (flagged; resolution pending).
A.4	Were clarification sessions conducted with all relevant stakeholder groups?	M1	S	BC-01 (07/02/2026): BioSec Academic Administrator, Parent Representative, Campus Operations Manager, Account Manager. TC-02 (09/02/2026): BioSec IT Manager, Network Infrastructure Lead. Both documented in AR §3.3–3.4.
A.5	Were meeting minutes formally recorded with attendees, focus, and outcomes?	M1	S	BC-01 and TC-02 documented in AR §3.3–3.4 with attendees, clarification focus, and key outcomes.
A.6	Were all compound requirements decomposed into atomic requirements?	M1, M2	S	AR §2.2–2.3 documents decomposition of BOR-1-MOB-02, BOR-1-MOB-03, BOR-1-WEB-01, BOR-1-WEB-02, TR-1-BE-01, and TR-1-BE-05 into atomic requirements with traceability.
A.7	Were open issues formally documented with resolution deadlines?	M1	PS	AR §3.5 documents three open issues with stated deadlines and assigned parties. However, the RPO/RTO deadline of 20/02/2026 was not met prior to SRS finalisation (see PA-ERR-001).

ID	Checklist Item	Method	Rating	Notes
A.8	Were models from the PS carried forward into the Analysis phase?	M1, M2	S	paper-napkin-model.jpg (original from PS) and context-diagram.pdf (revised) both present in repository requirement-models/.
A.9	Is there proof that the PS models were revised during Analysis?	M1, M2	S	context-diagram.pdf is a revised version reflecting TC-02 outcomes with six external entities. Original PS model retained separately. Data Dictionary revised with three new tables (C.7–C.9).
A.10	Does the IRC exist as a standalone, usable document?	M1	S	IRC_Nevon_Solutions.pdf is a standalone checklist-format document with 9 categories. Extractable and distributable.
A.11	Is there evidence that the IRC was actually used (not merely created)?	M1	S	IRC document contains 9 completed categories with project-specific findings tied to PS requirements. Not a blank template.
A.12	Did the vendor have a plan for the Analysing phase?	M1	S	BioSec Gantt Chart.xlsx documents structured sessions BC-01 and TC-02 with assigned dates and deliverables.
A.13	Were items from the PS properly codified in the repository?	M1, M2	S	bizops-requirements.md and technical-requirements.md carry a consistent BOR/TR identifier scheme. All requirements codified with unique IDs.
A.14	Was the biometric enrolment process elicited as part of the Analysing phase?	M1, M2	NS	No enrolment requirement appears in either the repository or the SRS. BC-01 and TC-02 outcomes make no reference to enrolment. See PA-ERR-004.

A.2 Domain B: Specifying Method

ID	Checklist Item	Method	Rating	Notes
B.1	Does the SRS reference a recognised template?	M1	S	srs_template-ieee.doc present in repository. SRS structure follows IEEE format.
B.2	Are all SRS requirements traceable to PS/repository requirements via iRTM?	M1, M2	PS	iRTM is present in SRS Appendix D. However, the absence of enrolment requirements (PA-ERR-004) means the iRTM cannot account for this process gap.

ID	Checklist Item	Method	Rating	Notes
B.3	Does the iRTM exist as a standalone, extractable file?	M1	S	iRTM.docx is a standalone .docx file in the repository. Not embedded in the SRS PDF.
B.4	Is terminology consistent across all SRS sections and artefacts?	M2	NS	Attendance status terminology is inconsistent across SRS sections. UR-MOB-01 uses "Attended, Absent, Late, or Excused"; §4.3.2 uses "Excused, Attended, Late, Absent"; Data Dictionary Table C.4 uses "Present." See PA-ERR-005.
B.5	Were all open issues from the Analysing phase resolved before SRS finalisation?	M1, M2	NS	All three open issues in AR §3.5 remain unresolved in the submitted SRS despite a stated deadline of 20/02/2026. See PA-ERR-001.
B.6	Were contradictory requirements identified during Clarify and resolved before Specifying?	M2	NS	The 2FA scope contradiction between UR-MOB-06 and NFR-SEC-03 was not resolved. BC-01 outcomes make no reference to 2FA scope. See PA-ERR-003.
B.7	If new models were created in Specifying, are the source files present?	M1	N/A	No new models created during Specifying phase. All models originated in Analysing and were carried forward.
B.8	If a Change Mitigation template is mentioned, does the template file exist?	M1	S	Change Request Form Template.docx present in repository requirement-models/.
B.9	Was the SRS submitted at a final ($\geq$ Version 1.0) version?	M1	NS	SRS bears Version 0.8. Analysing Report was released at Version 1.0. See PA-ERR-006.

A.3 Domain C: Validating Method

A.3.1 C.I Customer/Vendor Validation

ID	Checklist Item	Method	Rating	Notes
C.1	Was the SRS formally signed off by designated BioSec stakeholders before external audit release?	M3	NS	All four sign-off fields in SRS §9 are blank. No names, signatures, or dates recorded. See PA-ERR-002.

ID	Checklist Item	Method	Rating	Notes
C.2	Is there evidence of a spell check before SRS submission?	M3	PS	SRS Declaration page states the document was "reviewed, validated, and approved" by the Project Lead, but does not record spell check as a distinct activity. No standalone checklist or log found.
C.3	Is there evidence of a grammar check before SRS submission?	M3	PS	No documentary evidence of a grammar check was found.
C.4	Is there evidence of an editorial review before SRS submission?	M3	NS	No editorial check record found. Terminology inconsistencies (PA-ERR-005) indicate editorial review was either not performed or was ineffective.

A.3.2 C.II QC Audit Process

ID	Checklist Item	Method	Rating	Notes
C.5	Does the QC team have a documented audit plan?	M1	S	QC Report §3 contains a structured plan: target workproduct identified, context and goals stated, 10 inspection techniques listed, work breakdown with 4 team members. See Section 6.
C.6	Does the QC team have a structured checklist?	M1	S	QC Report §4 and Appendix B contain a self-contained checklist of 21 items (QC1–QC21) across 7 SRS sections. See Section 6.
C.7	Did the QC team document findings with evidence?	M1	S	QC Report §5–6 document 12 findings across three error types: Authorship (4), Omission (4), Contradiction (4). See Section 6.
C.8	Are QC findings saved in a retrievable report format?	M1	S	QC Report submitted as a standalone PDF with structured appendices. All 4 team members signed off (§8). See Section 6.

A.4 Domain D: Process Artefact Integrity

ID	Checklist Item	Method	Rating	Notes
D.1	Are all required process artefacts present?	M1	S	All artefacts listed in SRS Table of Contents and repository README confirmed present.

ID	Checklist Item	Method	Rating	Notes
D.2	Are process artefacts internally consistent with SRS content?	M2	PS	Context DFD documents six external entities consistent with SRS §2.1. However, Data Dictionary Table C.4 uses "Present" inconsistent with SRS §3.1 and §4.3.2.
D.3	Do all deliverables carry a Declaration of Use of AI Tools and a Declaration on Plagiarism and Originality?	M1, M3	S	Both the Analysing Report and the SRS carry both declarations, signed by Project Lead Loh Wen Xuan.
D.4	Is the requirements repository structured correctly with all required fields per requirement?	M1	S	Repository README confirms all seven fields. Sample review confirms compliance across all requirements.
D.5	Are Change Mitigation Strategies documented in the SRS?	M1	S	SRS §8 documents three strategies: Anticipatory Contingency Requirements, Open Issues, and High-Risk Change Control Process.
D.6	Is a Risk Analysis present in the Analysing Report?	M1	S	Nevon Risk Analysis Matrix.docx in repository. AR §5 documents risks with likelihood and impact ratings.
D.7	Are version control and document metadata complete across all deliverables?	M1	PS	AR (v1.0) has complete metadata. SRS (v0.8) has complete metadata but is below expected release threshold (PA-ERR-006).

B Detailed QA Findings: Evidence Tables

B.1 PA-FND-001: Evidence

Source	Location	Content
Analysing Report	§3.5	Three open issues; RPO/RTO deadline 20/02/2026
SRS	§2.5	Biometric template storage still open; kill date 15/03/2026
SRS	§3.3	MCS encryption handshake still an open issue
SRS	§6.5 NFR-DOC-03	Disaster recovery RPO/RTO "still pending BioSec approval"

Confirmation: The SRS date of 03/03/2026 post-dates the RPO/RTO resolution deadline of 20/02/2026. The SRS text itself acknowledges all three issues as unresolved. No resolution or deferral documentation was presented to the QA auditor.

B.2 PA-FND-002: Evidence

Source	Location	Content
SRS	§9 Sign-off	All four rows blank: Name, Signature, Date all empty
Repository	Full file listing	No signed declaration, confirmation document, or sign-off record found
Repository	Interview records	No record of verbal or alternative sign-off confirmation found
Analysing Report	Declaration, p.3	Correctly completed: Loh Wen Xuan, signature present, dated 13/02/2026

Confirmation: Direct inspection of SRS §9 confirms all sign-off fields are blank. Full repository inspection confirms no supplementary authorisation evidence exists. The Analysing Report's correctly completed sign-off confirms the team is familiar with the protocol. The Validating method was not completed.

B.3 PA-FND-003: Evidence

Source	Location	Content
SRS	§3.1 UR-MOB-06	2FA required for parents and teachers
SRS	§5.3 NFR-SEC-03	2FA required for administrators only
Analysing Report	§3.3 BC-01	No reference to 2FA scope
Analysing Report	§3.4 TC-02	No reference to 2FA scope

B.4 PA-FND-004: Evidence

Source	Location	Content
SRS	§4 System Features	No enrolment feature section; seven features, none covering enrolment
Repository	bizops-requirements.md	No enrolment requirement present
Repository	BOR-0-03 Acceptance Criteria	"Parental consent is collected before biometric enrolment" references enrolment without a corresponding requirement
Analysing Report	§3.3–3.4	No enrolment process in documented outcomes

B.5 PA-FND-005: Evidence

Source	Location	Content
SRS	§3.1 UR-MOB-01	"Attended, Absent, Late, or Excused"
SRS	§4.3.2	"Excused, Attended, Late, Absent"
SRS	Data Dictionary Table C.4	"Present" — diverges from both SRS §3.1 and §4.3.2

B.6 PA-FND-006: Evidence

Source	Location	Content
SRS	Cover page	Version 0.8, Date: 03 March 2026
Analysing Report	Cover page	Version 1.0, Date: 08 February 2026

C Detailed QC Process Audit

C.1 QC Plan Assessment

The QC Report §3 contains a documented audit plan with the following elements:

Plan Element	Status	Evidence
Target workproduct	Present	§3.1: NS-SRS-2026-001, Version 0.8, prepared by Nevon Solutions for BioSec Educational Institution
Context and goals	Present	§3.2: Verifies SRS clarity, completeness, and internal consistency
Inspection techniques	Present	§3.4: 10 techniques including close document inspection, perspective-based reading, cross-section contradiction checking, codification verification, model-to-text consistency, omission mapping, scenario-based reading, assumption reading, fact-checking, and spellcheck/grammar review
Work breakdown	Present	§3.5/Appendix A Table A-1: Four roles: QC Audit Lead, Requirements Reviewer, Traceability Reviewer, Editorial and Structure Reviewer
Timeline	Present	§3.6/Appendix A Table A-2: Four stages totalling ∼6 hours
Reporting protocol	Present	§3.7: Findings recorded in report; audit team does not amend the SRS

C.2 QC Checklist Assessment

21 items (QC1–QC21) across seven SRS sections: Introduction (QC1–3), Glossary (QC4–6), Overall Description (QC7–9), Functional Requirements (QC10–12), Interface Requirements (QC13–15), Non-Functional Requirements (QC16–18), Addenda (QC19–21). Items are CRaM-specific to the Biometric Student Attendance System and directly address all three QC error families.

C.3 QC Error Detection Assessment

The QC team documented findings covering all required error families: Contradiction, Omission, and Authorship.

C.4 QC Findings Assessment

Findings were documented with Context-Result-Analysis structure and severity classifications. All findings are signed off.

C.5 QC Sign-off and Addenda

All four team members signed. Addenda include Glossary (10 terms), Appendix A (Plan Tables), Appendix B (Checklist), Appendix C (Error Detection), Appendix D (Findings Summary), and References (5 sources). All appendix content is correctly linked from the main report body.

D Audit Work Allocation and Timeline

D.1 Audit Timeline

Stage	Date	Activities	Hours
Stage 1	15 Mar 2026	Document collection; confirm access; prepare audit checklist	2
Stage 2	16 Mar 2026	Review all artefacts; apply checklist Domains A and B	2
Stage 3	17 Mar 2026	Apply checklist Domains C and D; request supplementary evidence; document non-compliance	2
Stage 4	18 Mar 2026	CRaM analysis; cross-reference QC Report; consolidate evidence	1
Stage 5	19 Mar 2026	Report drafting, review, and finalisation for Nevon Solutions submission	1
Total			**8**

D.2 Artefacts Reviewed

Document ID	Title	Version	Date
BS-PS-2026-001	BioSec Project Specification	1.0	—
NS-AR-2026-001	Nevon Solutions Analysing Report	1.0	08/02/2026
NS-SRS-2026-001	Nevon Solutions SRS	0.8	03/03/2026
NS-IRC-2026-001	Incongruous Requirements Checklist	1.0	—
—	iRTM (.docx), Gantt Chart (.xlsx), Change Request Form (.docx), IEEE SRS Template (.doc), Risk Analysis Matrix (.docx)	—	—
VL-QCR-2026-001	Veridion Labs QC Audit Report	1.0	—

E Glossary

Term	**Definition**
QA	Quality Assurance. Auditing the RE *process* for compliance with defined process requirements.
QC	Quality Control. Inspecting a *work product* (e.g., the SRS) for authorship defects, omissions, and contradictions.
RE	Requirements Engineering. The structured set of methods used to elicit, analyse, specify, and validate software requirements.
Analysing Method	BCIC process (Breakdown, Clarify, Interpret, Categorise) applied to the PS to produce a regulated-revision requirements repository.
Specifying Method	Interpret, Categorise, and Specify phases applied to the requirements repository to produce the SRS.
Validating Method	The fourth RE method. Applies QC measures onto the SRS and QA audit of the vendor's RE conduct.
Process Non-Compliance	The sole QA error type. An instance where the RE process deviated from defined requirements.
CRaM	Crisp, Realistic and Measurable. Quality criterion for requirements and audit findings.
SRS	Software Requirements Specification. Primary work product of the Specifying method.
PS	Project Specification. Customer's source document for requirements.
AR	Analysing Report. Produced following BCIC-based regulated revision of the PS.
IRC	Incongruous Requirements Checklist. Tool for the Breakdown phase. Must be standalone and distributable.
iRTM	Inceptive Requirements Traceability Matrix. Chain-of-custody from PS to SRS. Must be a standalone file.
BCIC	Breakdown, Clarify, Interpret, Categorise. Four core activities of the RE Analysing method.
Open Issue	Specification matter unresolved during clarification, documented with deadline for resolution before SRS finalisation.
PDPA	Personal Data Protection Act (Singapore, 2012, as amended). Governs biometric and attendance data in this project.
Annex	Physical copy of a standalone external document included for reference. Distinguished from Appendices (lettered).
Declaration	Signed legal statement confirming the truth of a claim. Used as halfway evidence when primary documentation is absent.

F References

Ref.	Document	Notes
[1]	BS-PS-2026-001: BioSec Project Specification, Version 1.0	Customer source document
[2]	NS-AR-2026-001: Nevon Solutions Analysing Report, Version 1.0, 08 February 2026	Primary artefact for Analysing method audit
[3]	NS-SRS-2026-001: Nevon Solutions SRS, Version 0.8, 03 March 2026	Primary artefact for Specifying and Validating audit
[4]	VL-QCR-2026-001: Veridion Labs QC Audit Report	Cross-referenced for corroboration
[5]	IEEE Std 830-1998: IEEE Recommended Practice for SRS	SRS template basis
[6]	ICT2113 Software Modelling & Analysis, L08: Validating, Prof Kevin Anthony Jones, 25/12/2025	Course reference for QA/QC methodology

Annex

A Evidence Screenshots

This annex contains supporting screenshots and visual evidence linked to the QA findings documented in Section 5. Each subsection corresponds to a finding and presents relevant document excerpts, requirement statements, or process artefacts that substantiate the non-compliance instance.

A.1 PA-FND-001: Open Issues Unresolved at SRS Finalisation

Related Finding: Section 5, PA-FND-001

Screenshots:

3.5 Open Issues for SRS Documentation

The following technical specifications required additional investigation beyond clarification session scope and were agreed to be documented as open issues in the initial SRS:

- **MCS Integration Security:** Specific encryption handshake protocol for MCS API communication requires review of complete MCS technical documentation not available during clarification sessions. Nevon Solutions to coordinate with MCS vendor for security protocol specifications.
- **Disaster Recovery RPO/RTO:** BioSec requires additional internal review of acceptable Recovery Point Objective and Recovery Time Objective for attendance data. Stakeholders to provide disaster recovery requirements by 20/02/2026 for incorporation into SRS.
- **Biometric Template Storage Format:** Confirmation needed on whether biometric templates must conform to ISO/IEC 19794-2 format for potential future interoperability or can use proprietary sensor format for optimized performance. Decision impacts template storage size and matching algorithm performance.

Annex 1: Analysing Report §3.5: Open Issues table with deadlines

Issues List

This appendix consolidates all unresolved TBD (to be determined) references that remain in this SRS. These correspond to the open issues defined under Change Mitigation Strategy CM#2 (Section 8.2). Full details including descriptions, fallback actions, rationale, and kill dates are provided in Table 2 of Section 8.2.

Issue ID	Issue Title	Status	Kill Date	SRS Reference(s)
CM-OI-01	MCS Integration Security	Open	20 Mar 2026	Section 2.7; Section 3.3; FR-SW-01; FR-SW-02; NFR-SEC-05
CM-OI-02	Disaster Recovery RPO/RTO	Open	20 Mar 2026	Section 2.7; NFR-DB-01; NFR-DOC-03
CM-OI-03	Biometric Template Format	Open	15 Mar 2026	Section 2.5; Section 3.2; NFR-DB-03

Table B.1 – Open Issues Summary

Annex 2: SRS §2.5, §3.3, §6.5: Unresolved open issues (biometric template, MCS encryption, RPO/RTO)

A.2 PA-FND-002: SRS Sign-off Entirely Absent

Related Finding: Section 5, PA-FND-002

Screenshots:

9. SRS Sign off

This Software Requirements Specification (NS-SRS-2026-001) has been reviewed and is submitted for approval. The document defines the agreed baseline of functional and non-functional requirements for the Biometric Student Attendance System, and shall serve as the authoritative reference for all subsequent design, development, testing, and deployment activities.

By signing below, the Project Lead certifies the following:

- The Software Requirements Specification (Version 0.8) has been prepared in accordance with the project's requirements engineering methodology and is internally consistent.
- All requirements derived from the Project Specification (BS-PS-2026-001) have been accounted for through the Analysing process and are traceable via the Inceptive Requirements Traceability Matrix (iRTM).
- The documented requirements constitute the approved development baseline, inclusive of open issues and change mitigation strategies defined herein.
- All modifications to this baseline after sign-off shall be governed exclusively by the Change Mitigation Strategies specified in Section 8 of this document.

Vendor Authenticator

Loh Wen Xuan		06/03/2026
Name (Project Lead)	**Signature**	**Date**

Designated Approvers – Vendor (Nevon Solutions)

Role	Name	Signature	Date
Technical Lead	James Koh		
System Architect Lead	Rachel Lee		

Designated Approvers – Customer (BioSec)

Role	Name	Signature	Date
Project Manager	Marcus Tan Wei Liang		
Head of Operations	Daniel Rodriguez		

Annex 3: SRS §9: Blank sign-off fields (all four rows empty)

6. Sign Off

This section certifies that the Biometric Student Attendance System Analysing Report has been examined and approved by stakeholders. By signing below, the Project Lead and designated approvers confirm that:

- All requirements from the project specification have been systematically evaluated using the Incongruous Requirements Checklist and verified internally.
- This document establishes a validated foundation for proceeding to the subsequent Software Requirements Specification development.
- The document is authorised for distribution as a component of the bidding or tender procedure.

Role	Name	Signature	Date
Project Lead	Loh Wen Xuan		13/2/2026
Designated Approvers			
Technical & Compliance Analyst	Chu Wee How Brandon		13/2/2026
Requirements Manager	Ang Ke Ying		13/2/2026
Quality Assurance Analyst	Lim Zong Wei Glenn	Glenn	13/2/2026
Solutions Architect	Haimirul Hakim Bin Hamidi		13/2/2026

Annex 4: Analysing Report: Correctly completed sign-off (reference)

A.3 PA-FND-003: 2FA Scope Contradiction Not Resolved

Related Finding: Section 5, PA-FND-003

Screenshots:

<u>**Parent and Teacher Mobile Application Interface Requirements**</u>

- **UR-MOB-01:** The parent interface shall display monthly attendance records for the parent's registered child only, with attendance status shown as Attended, Absent, Late, or Excused.
- **UR-MOB-02:** The parent interface shall provide a leave request submission screen that allows date selection, reason entry, and optional medical certificate upload for MCS validation.
- **UR-MOB-03:** The teacher interface shall display attendance records for assigned classes only, in a read-only format with no approval or submission functionality.
- **UR-MOB-04:** The mobile application shall display push notifications for absence events and leave status updates.
- **UR-MOB-05:** Primary tasks, including attendance viewing and leave request submission, shall be reachable within three interactions from the home screen, to accommodate users with varying technical literacy.
- **UR-MOB-06:** The parent and teacher authentication shall include a 2-Factor Authentication screen process.

Annex 5: SRS §3.1 UR-MOB-06: 2FA requirement for parents and teachers

5.3 Security Requirements

The security requirements protect biometric and personal data processed by the system. All requirements are consistent with Singapore's Personal Data Protection Act (PDPA) 2012, as amended. The system handles biometric data from minors, which necessitates the highest level of data security.

- **NFR-SEC-01:** The system shall encrypt biometric data, including fingerprint templates, at rest using AES-256 and in transit using TLS 1.2 or higher. No plaintext biometric data shall be stored or transmitted at any point. (TR-1-BE-03, TR-1-BE-05b)
- **NFR-SEC-02:** The system shall process and store all PII, including student records, parent contact information, and attendance data, in accordance with PDPA retention policy. (TR-1-BE-03)
- **NFR-SEC-03:** The system shall enforce two-factor authentication (2FA) for all administrator accounts accessing the web portal. (TR-1-BE-03; TR-1-CPR-01)
- **NFR-SEC-04:** The system shall lock all user accounts after a configurable number of failed login attempts, with a configurable lockout duration. All lockout events are logged with timestamp and account identifier. (TR-1-CPR-01)
- **NFR-SEC-05:** The system shall conduct all MCS API communications using HTTPS and OAuth 2.0 authentication. No other authentication mechanism shall be permitted. (TR-1-BE-02)
- **NFR-SEC-06**: The system shall retain audit logs, including login events, attendance changes, leave approvals, and access denials, for at least 12 months and shall protect them from modification or deletion. (TR-1-CPR-01)
- **NFR-SEC-07:** The system shall restrict web portal access to campus networks or secure VPN connections. Direct internet access to the administrator portal shall not be permitted. (TR-1-CL-02)
- **NFR-SEC-08:** The system shall require mobile applications to be code-signed with BioSec institutional certificates before distribution on Google Play and Apple App Store. (TR-1-CL-01d)
- **NFR-SEC-09:** The system shall encrypt all locally cached data on mobile devices using AES-256. Data cached for offline use shall be automatically purged after 48 hours. (TR-1-CL-01f)

Annex 6: SRS §5.3 NFR-SEC-03: 2FA requirement for administrators only

Meeting Code: BC-01	**Date: 07/02/2026**
Attendees: Loh Wen Xuan (Business Analyst), Dr. Catherine Lim (BioSec Academic Administrator), Ms. Sarah Tan (Parent Representative), Mr. David Wong (Campus Operations Manager), Ms. Florence (Account Manager), Lim Zong Wei Glenn (Quality Assurance Analyst), Ang Ke Ying (Requirements Manager)	

Clarification Focus: Resolution of business operations requirement ambiguities and correction of structural and imperative issues
Key Outcomes: • Biometric verification quantified as 2 seconds processing time supporting 50 students per device per 10 minutes • After 3 failed retries, manual attendance authorized with separate flagging • Fixed-location deployment selected at building entrances and main corridors • Data synchronization defined as 5-minute sync with 30-minute maximum staleness • Leave submission validates mandatory fields with field-level errors, save-as-draft preserves incomplete submissions for 7 days • Notification delivery target repositioned to 98% within 15 minutes with SMS fallback after 30 minutes • Compliance thresholds configured by administrators at 80% default attendance per term • Leave requests queue during MCS outage with manual override capability, requests pending beyond 4 hours flagged for manual review

Annex 7: Analysing Report §3.3: BC-01 clarification session outcomes (no 2FA resolution)

Meeting Code: TC-02	**Date: 09/02/2026**
Attendees: Loh Wen Xuan (Business Analyst), Mr. James Koh (BioSec IT Manager), Ms. Rachel Lee (Network Infrastructure Lead), Ms. Florence (Account Manager), Chu Wee How Brandon (Technical & Compliance Analyst), Lim Zong Wei Glenn (Quality Assurance Analyst), Haimirul Hakim Bin Hamidi (Solutions Architect), Ang Ke Ying (Requirements Manager)	
Clarification Focus: Validation of technical implementation boundaries and removal of premature design specifications	
Key Outcomes: • FAR target revised to 0.01% based on commercial sensor capabilities in high-traffic educational environments • Peak load specified as minimum 1,500 concurrent users with graceful degradation to 2,000 at 4-second response, auto-scaling triggers at 70% CPU • MCS API confirmed as HTTPS REST with OAuth 2.0, 5-second timeout, 3 retry attempts with exponential backoff, 99.5% SLA • Three-tier issue severity: Critical with 4-hour response and 24-hour resolution, High with 8-hour response and 48-hour resolution, Medium with 24-hour response and 5-day resolution • Offline mode applies only to attendance viewing and leave request drafting, 48-hour maximum offline duration, 50MB local storage limit • Context DFD updated to reflect all external system interfaces identified during clarification. (see Fig. 2 in Appendix B) • Data Dictionary updated with new tables for NotificationLog, BiometricScan, and ExternalSystemLog to support external system integrations and clarified requirements. (see Table C.7 to C.9 in Appendix C)	

Annex 8: Analysing Report §3.4: TC-02 clarification session outcomes (no 2FA resolution)

A.4 PA-FND-006: SRS Submitted Below Final Version Threshold

Related Finding: Section 5, PA-FND-006

Screenshots:

NEVON SOLUTIONS
Information Technology

Software Requirements Specification

Biometric Student Attendance System

Date: *3 March 2026*
Version: *0.8*
Document ID: *NS-SRS-2026-001*

Prepared by: *Nevon Solutions Business Analysis Department*

Name	Role / Designation
Loh Wen Xuan	Project Lead
Ang Ke Ying	Requirements Manager
Chu Wee How Brandon	Technical & Compliance Analyst
Haimirul Hakim Bin Hamidi	Solutions Architect
Lim Zong Wei Glenn	Quality Assurance Analyst

Annex 9: SRS cover page: Version 0.8, 03 March 2026

NEVON SOLUTIONS
Information Technology

Analysing Report

Biometric Student Attendance System

Date: *8 February 2026*
Version: *1.0*
Document ID: *NS-AR-2026-001*

Prepared by: *Nevon Solutions Business Analysis Department*

Name	**Role / Designation**
Loh Wen Xuan	Project Lead
Ang Ke Ying	Requirements Manager
Chu Wee How Brandon	Technical & Compliance Analyst
Haimirul Hakim Bin Hamidi	Solutions Architect
Lim Zong Wei Glenn	Quality Assurance Analyst

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Annex 10: Analysing Report cover page: Version 1.0, 08 February 2026

End of PA-QAR-2026-001